

JBG060: Q&A 2

Flood dynamics in South Sudan

ZOA International & Zero Hunger Lab | Janne de Rooij & Joris Wagenaar | September 21st, 2026



Clarify starting point

- No experience yet with AA
- No experience working with data yet either
 - a) **Understand risks** associated with **data** used
 - b) Help us **understand flood dynamics** better, such that we can **inform decision making** based on this
- Lot of questions which we don't have the answer to:
instead, **your analysis** should **help us** in **answering** those

ZOAs way of working

- **Locally-led approach**

- Local decisions > country-wide assessment
- Work together with local communities
- All decisions made at spatial scale of **community**

- Target people most at need

- Projects active in certain places, but since goal is to target those most need it, also relevant to know context in other locations
 - Along the river 'Lol' in Northern Bahr El Ghazal.
 - Flood-prone areas in Bor South county

How is such an AA plan designed?

- Local ZOA staff writes **AA proposal** based on collaboration with local community: their input defines proposal
 - Needs?
 - Impact?
 - How to reduce?
- ZOA HQ needs to **approve**
- Proposal submitted to a **project call** → funding received if ‘won’
- **Execution:**
 - Local ZOA staff: monitor early warnings.
 - Community involved in execution of AAs
 - No decision taken solely based on a model
 - Temporary presence of ZOA at location: local community should understand

Goal of project

- In order to decide **what** anticipatory action **where** at **what condition**, **first important to understand**
 - What can cause a flood to occur here?
 - What is the specific impact of this flood locally?
 - How do flood drivers differ per impact?
 - At what time scale can we predict this to occur?
 - What is the reliability of our prediction?
 - Then: able to decide which actions are initiated when, where
- your goal: **improve understanding**.
- **Rigorous** answer small sub-problem > **superficial** answer full problem
 - All findings should be **transparent**: why this finding, at what certainty?

What impact is ‘most important’?

- **Varies** per location.
- Priority impact described in active region: harvest failed due to flooding
- But ... does not make it only factor of interest

→ in order to define a ‘fair’ project, important to understand impact on all individuals and contexts

Small-holder vs large-scale farmers:

In the current rural areas, there are mainly small-holder farms.

Targeted households?

Households with **high exposure** to anticipated hazard + **limited capacity** to take protective action

- Female- and child-headed HH
- HH with disabilities or limited mobility/support
- HH with pregnant/lactating women
- HH with displaced and returnees, with limited assets/social support
- HH with limited sources of income/depleted food reserves

Anticipatory actions

- Preference of which action?
 - **Depends on situation**, according to exposure/vulnerability local population and the anticipated hazard impact
- Which action is required when?
 - **Depends** on anticipated impact
 - Action should only be included if **feasible to mitigate impact** [ex: migrating cattle]
- Should our model decide which action to initiate?
 - Not necessarily, it should **help inform which action is needed**
- Required implementation time: the longer, the better:
 - 14 days: definitely enough – robust upper bound
 - 10 days: suffices
 - 3 days: minimum

What are 'disappointing' outcomes

- Not improving our understanding of flood dynamics in South Sudan
- Providing non-interpretable or non-nuanced results
 - How certain are you?
 - Potential consequences if not accurate?
- Information that is not relevant for local decision-making

Consequences misses and false alarms

- Vulnerable people did not receive critical assistance on time
- Critical resources spend that could have been used to address **urgent needs**
- Decreased trust in early warning system
- False alarm: effort made by local communities to prepare for flood
 - missed income due to time,
 - reduced yield due to early harvesting
 - ...

Trade-off false alarms and misses

- **You will not provide a 100% reliable forecast, warning or recommendation**
- Aim: be able to transfer the reliability and risk associated with your recommendation
- Trade-off: **no one correct answer**
 - Provide us the knowledge such that we can decide this trade-off better

Specific questions

What information is **most important** to consider **before recommending livestock movement** away from flood impacted area?

Pre-requisite, based on local assessment: cattle can be moved to a safe location

Information before acting: inundation prediction - no access to food and risk of water-borne diseases

How are cattle relocation decisions organized during floods usually?

- **Decided locally:** community-broad advice, decision taken on household-level
- Unknown based on what

Are cash contributions a good solution for attempting to compensate the damages done to cropland?

Yes

Alternatives that directly supports recovery also good.

History of using nitrogen fertilizer?

No experience: could be applicable

Given a flood, we are estimating the crop damage. Good approach?

Yes: this is relevant

Our research objective can be implemented if combined with flood prediction. Is this sufficient?

Yes: this improves our knowledge

Focusing on independence and food security, should we focus on specific crop types more than other crop types?

Yes that is okay: if it makes your analysis more informative

Would you prefer a (machine learning) prediction model or a formula?

Both can work.

However, simple and interpretable is better than complex and black-box

All findings/recommendations need to be understood

What is 'crop damage'?

Multiple dimensions:

- Yield loss
- Nutritional loss
- Local food availability

Most relevant: **impact-based**

harvest loss per household in kg

change in estimated food insecurity

Would a highly detailed pixel-level prediction actually be useful or is that overly complicated?

The more detailed, the better, since this informs better decision making at community level

We want to evaluate our estimated crop loss against real data, where should we look for a dataset about crop damage?

We don't know this: recommend academic literature and FAO as starting point

What information would ZOA need from the model to make the decision to initiate an action if they receive a prediction that a flood will cause severe crop damage?

What damage is expected to take place where, when, and how certain are you?

Would it be important that local organizations can operate and interpret the system without depending on external technical organizations? What organizations would that be?

Yes.

For sure: local ZOA staff,

Ideally: possible to transfer this knowledge to local community leaders.

How do aid resources arrive to a region (cars, ships, planes) which routes do they usually take?

Depends on post-flood assessment

Always sourced as locally as possible: stimulates economy

Not a large-scale humanitarian operation such as WFP

instead: focus is last-mile, distribute via roads or boats

Do you consider explaining or predicting flood duration a useful/equally useful insight compared to flood onset predictions?

Yes: duration of inundation is important to understand a floods impact

How are aid resources currently allocated? What principles? Population/Consequence severity//Equally, etc. if there is another specific way?

Based on highest needs:

- Coping capacity
- Vulnerability
- Exposure to hazard

From ZHL's perspective, what would be the most valuable outcome of our project

1. understanding the conditions associated with flooding,
2. identifying potential flood precursors, or
3. investigating whether these precursors could provide useful early warning?

All relevant, but rigorous > superficial, hence likely 1 → 2 → 3

When is a result **practically useful**? Would identifying potential flood precursors or estimating their possible warning time already be valuable, **even if** we cannot develop a reliable prediction model?

Yes: don't expect you to solve the full problem.

Smaller sub-problem rigorously >>> full problem superficially

Would investigating upstream hydrological conditions, such as lake levels or river discharge, be a meaningful direction for the project

Yes: these are interesting flood drivers with long lead time!

What type of trigger releases your anticipatory funds: a **hard physical cutoff** (such as, more than X hectares submerged) or a **probability threshold** (such as, greater than a 70% chance of severe flooding)?

As long as it is a pre-determined condition being reached.

Potentially: combination.

“with at least $x\%$ certainty, flood will take place which inundates cropland resulting in at least $y\%$ of total yield reduction”

If you could receive one reliable piece of information 14 days before a major flood, what information would most influence your decision to prepare a cash transfer?

How are you expecting a flood to affect the local population in terms of inundation duration and extent, as well as the effect on crop yield and access to critical infrastructure, with what reliability?

According to the dataset, South Sudan seems to have relatively little arable land — but the lecture said flood destruction of harvests is a major problem. Could you help us understand this apparent discrepancy? For example, might the dataset not fully capture cultivated or seasonally used farmland?

Fact: approximately 80% of rural communities in South Sudan list agriculture as primary source of livelihood

So yes, the dataset might not capture agricultural land fully

What do you do with road information today (route convoys, choose where to pre-position, decide where to open a site?) Could you please give an example on how road access changes your decision-making.

Informs the vulnerability and coping capacity of local communities.

As a result, affects whether they are targeted to receive assistance or not.

What do you most value having access to (need assessment)? Given the food security mission, would e.g. distribution points or your own warehouses be the primary focus, or would healthcare facilities also be important?

- No warehouses or distribution points
- Idea: act in *anticipation of flood*
 - access to health influences peoples vulnerability & coping capacity during flood
 - Primary concerns: food security, livelihood preservation & health

The Logistics Cluster publishes access constraint maps for South Sudan. How do you work with the information, What do those not tell you that you wish they did?

No experience

Do you already use displacement data in your operations? How does population distribution and displacement across the country impact decision-making? Given South Sudan's complex ethnic groups and internal conflict, how is this handled?

- Humanitarian organizations are **politically/conflict neutral**
- Displacement +conflict data influences vulnerability and exposure of local population
- Conflict influences feasibility of which type of assistance is feasible to provide as well as what is needed

Does your programming target IDPs, host communities, pastoralists, or all of them?

All

What has ZHL specifically ruled out already in South Sudan?

Nothing

What rules would a trigger have to satisfy to be a viable trigger to consider in anticipatory action?

- Transparent
- Interpretable

Are there any areas that ZHL would not operate in?

Yes: if this does not reflect in addressing the most critical needs

Would it be useful for you to have multiple instances of our solution for the different types of AA based on their risk and time frame (and maybe other factors)?

For example having a trigger for cash distribution which activates when there is a 60% chance of a flood in 7 days, and a different trigger for early harvesting which activates when there is a 95% chance of a flood in 2 days.

Yes

Would it be useful to keep repeat floods in mind when designing our solution?

Yes